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ORIGINAL RESEARCH

Patellofemoral Joint Loading in Forward Lunge With Step Length and Height Variations

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The objective was to assess how patellofemoral loads (joint force and stress) change while lunging with step length and step height variations. Sixteen participants performed a forward lunge using short and long steps at ground level and up to a 10-cm platform. Electromyography, ground reaction force, and 3D motion were captured, and patellofemoral loads were calculated as a function of knee angle. Repeated-measures 2-way analysis of variance ($P < .05$) was employed. Patellofemoral loads in the lead knee were greater with long step at the beginning of landing (10°–30° knee angle) and the end of pushoff (10°–40°) and greater with short step during the deep knee flexion portion of the lunge (50°–100°). Patellofemoral loads were greater at ground level than 10-cm platform during lunge descent (50°–100°) and lunge ascent (40°–70°). Patellofemoral loads generally increased as knee flexion increased and decreased as knee flexion decreased. To gradually increase patellofemoral loads, perform forward lunge in the following sequence: (1) minimal knee flexion (0°–30°), (2) moderate knee flexion (0°–60°), (3) long step and deep knee flexion (0°–100°) up to a 10-cm platform, and (4) long step and deep knee flexion (0°–100°) at ground level.

Keywords: lunging, knee, force, stress

Forward lunges are performed functionally during activities of daily living and also when playing sports. They have been shown to be effective in recruiting and strengthening hip and thigh musculature muscles.[1–6] Hip strengthening, especially the gluteal muscles, is important to control frontal and transverse plane movements of the femur, which is half of the patellofemoral joint. Thigh strengthening, especially the quadriceps, helps properly control the patella, which is the other half of the patellofemoral joint. Exercises like the forward lunge that strengthen both hip and thigh musculature are important to enhance patellofemoral joint stability and improve optimal interaction between the femur and patella during activity. Clinicians commonly employ forward lunge exercises for the rehabilitation of individuals with knee injuries and pathologies, including patients with patellofemoral pain syndrome (PFPS).[7–11]

High patellofemoral joint force (and stress, which is force divided by contact area) may result in PFPS from numerous soft tissues, such as synovial plicae, infrapatellar fat pad, retinacula, joint capsule, and patellofemoral ligaments.[12] High patellofemoral joint force can also elevate subchondral bone stress in the patellofemoral joint.[13] Because the subchondral bone plate is rich in pain receptors,[14] increased subchondral bone stress may also result in PFPS.[12] Patellofemoral joint stress can result in cartilage degeneration and a decrease in the ability of the cartilage to distribute patellofemoral force.[13]

The goal of patellofemoral rehabilitation is to initially strengthen hip and thigh muscles with minimum patellofemoral joint loading, gradually progress patellofemoral loading over time, and create dynamic knee joint control to assist in patella tracking.[15,16] In patellofemoral rehabilitation progression, the forward lunge is initially performed with no external resistance (bodyweight only) and progressed to using weights (dumbbells or barbells) or other external resistance, such as resistance bands, and this progression increases both hip and thigh muscle recruitment and patellofemoral force (and stress).[5,17,18] Moreover, forward lunge exercises are commonly performed and progressed in knee rehabilitation settings using different step lengths (short step vs long step) and step heights (ground level vs elevated platform). Three previous studies quantified patellofemoral force (and stress) during forward lunge exercises,[17–19] but it is currently unknown how patellofemoral force (and stress) change while performing the bodyweight forward lunge with step length and step height variations. Therefore, the purpose of this study was to quantify patellofemoral force (and stress) while performing the bodyweight forward lunge with step length variations (short step vs long step) and step height variations (ground level vs 10-cm platform). Our hypotheses are that patellofemoral force (and stress) will be significantly greater with a short step and at ground level.

Methods

Sixteen healthy adults (8 men and 8 women) without a history of patellofemoral pathology participated. The mean (SD) age, mass, and height for the male participants were 28.9 (7.9) years, 77.3 (6.6) kg, and 175.9 (2.3) cm, respectively, whereas the women were 30.6 (9.8) years, 61.2 (6.8) kg, and 166.4 (8.5) cm, respectively. To help optimize the quality of the electromyographic (EMG) signal, all participants had average or below average body fat, which was assessed by Baseline skinfold calipers (model

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68900, Country Technology, Inc) and appropriate regression equations and body fat standards set by the American College of Sports Medicine.[20] Mean (SD) body fat was 9.0% (3.9%) for men and 18.7% (3.3%) for women. This study was approved by the institutional review board at California State University (Sacramento).

All subjects reported to the Biomechanics Laboratory 1 week prior to data collection for a pretest session. As it has been demonstrated that trunk position can affect hip muscle activity as well as patellofemoral force (and stress),[4,19] during the pretest, each participant's forward trunk position at the lowest position of the lunge was screened. As previously reported,[17,18] most individuals skilled in lunge exercises achieve between 0° and 20° of forward trunk tilt (0° represented the trunk being in the vertical position) at the lowest position of the lunge descent. Therefore, only participants who met this 0° to 20° forward trunk tilt criterion during the pretest session (measured with a standard bubble goniometer by the lead author, Escamilla) were allowed to participate in data collection the following week.

Also during the pretest session, the subject's forward lunge with a long step and short step were determined and practiced. The forward lunge with a long step was determined by using a step length that resulted in the right leg (shank) being approximately vertical at the lowest position of the lunge, thus maintaining the knee over the ankle joint (Figure 1B and 1D). The mean (SD) step length (measured from left toe to right heel at the lowest position of the lunge) for the forward lunge with a long step was 90.3 (4.9) cm for men and 81.3 (5.8) cm for women. The step length for the forward lunge with a short step was, by convention, half the distance used for the forward lunge with a long step, thus causing the anterior surface of the right knee to translate forward beyond the toes of the right foot (Figure 1A and 1C). These step length measurements for the forward lunge were subsequently used the following week during data collection. Each participant then practiced all step length and step height (landing at ground level vs landing up to a 10-cm height platform—Figure 1C and 1D) lunge variations. With both feet together, the subject lunged forward, and after lead foot (right foot) contact with a force platform at ground level or up to a 10-cm height force platform,

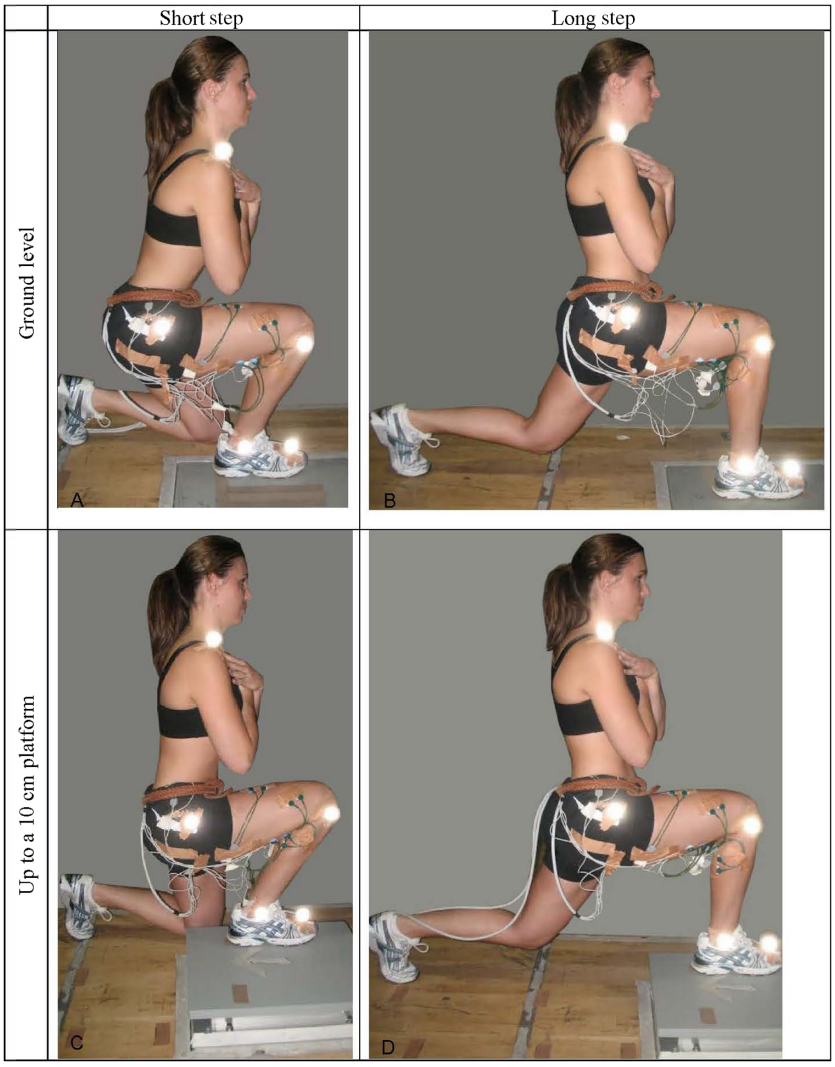


Figure 1 — Forward lunge with (A) a short step at ground level, (B) a long step at ground level, (C) a short step up to a 10-cm platform, and (D) a long step up to a 10-cm platform.

the right knee slowly flexed until maximum knee flexion (approximately 100°) was obtained as the trail knee (left knee) lightly made contact with the ground, completing the lunge descent. From this position, the subject immediately pushed backward off the force platform straightening the knee (lunge ascent) and returned to the upright standing position with both feet together. A metronome set at 60 beats per second was used to help ensure the right knee flexed and extended at approximately 45°/s to 50°/s, with 2 seconds during the descent (2 beats) and 2 seconds during the ascent (2 beats), and the metronome was used during data collection as well. Each participant was able to perform all lunge variations pain free and with proper technique, as assessed by the lead author (Escamilla), who is a certified strength and conditioning specialist and a licensed physical therapist.

Each subject reported back to the Biomechanics Laboratory 1 week after the pretest session for data collection. Blue Sensor (Ambu Inc) disposable oval-shaped surface electrodes (22-mm wide and 30-mm long—type M 00 S) were used to collect EMG data and were placed in a bipolar configuration along the longitudinal axis of each muscle with approximately 3 cm between the centers of the electrodes. Prior to applying the electrodes, the skin was prepared by shaving, abrading, and cleaning with isopropyl alcohol wipes to reduce skin impedance. Electrode pairs were then placed on the subject's right side using previously described locations[17,18,21,22] for the following muscles: (1) rectus femoris, (2) vastus lateralis, (3) vastus medialis, (4) medial hamstrings (semimembranosus and semitendinosus), (5) lateral hamstrings (biceps femoris), and (6) gastrocnemius (middle portion between medial and lateral bellies).

Reflective markers (3.8 cm in diameter) were attached to the following bony landmarks, as previously described[17,18,22]: (1) third metatarsal head of the right foot, (2) medial and lateral malleoli of the right leg, (3) upper edges of the medial and lateral tibial plateaus of the right knee, (4) posterosuperior greater trochanters of the left and right femurs, and (5) lateral acromion of the right shoulder.

Motion of the reflective markers was tracked with an 8-camera motion analysis system (Vicon Peak Performance Technologies, Inc.) with a sampling rate of 60 Hz. Force data were collected at 960 Hz using an AMTI force platform (model OR6 6 2000, Advanced Mechanical Technologies, Inc). EMG data were collected at 960 Hz using a Noraxon Myosystem unit (Noraxon USA, Inc). The EMG amplifier bandwidth frequency was 10 to 500 Hz with an input impedance of 20,000 kΩ and a common mode rejection ratio of 130 dB. Video, EMG, and force data were electronically synchronized and simultaneously collected.

Once the electrodes and reflective markers were positioned, the subject performed a 5- to 10-minute dynamic warm-up consisting of the toe walk, heel walk, knee to chest walk, walking quad stretch, straight-leg march, inverted hamstring stretch, inchworm, power skip, rotational lunge, lateral lunge, and spiderman lunge. The subject then practiced performing a few repetitions for each of the 4 lunge variations as described in the pretest session. In a randomized order, each participant then performed 3 repetitions of the 4 forward lunge variations using bodyweight only (Figure 1A–1D) as force platform, EMG, and video data were collected.

Subsequent to completing all 4 sets of lunge variations, EMG data were collected during maximum voluntary isometric contractions (MVICs), as previously described.[17,18,22] The MVICs for the rectus femoris, vastus lateralis, and vastus medialis were collected in a seated position at 90° knee and hip flexion during a maximum effort knee extension. The MVICs for the lateral and medial hamstrings were collected in the same seated position during a maximum effort knee flexion. MVIC for the gastrocnemius was collected during a maximum effort standing unilateral stance toe raise with the ankle positioned approximately halfway between neutral and full plantar flexion. Two 5-second trials were collected for MVIC of each muscle.

A fourth-order Butterworth low-pass filter was used to smooth the data with a cutoff frequency of 6 Hz for the 3-dimensional motion data and 12 Hz for the force platform data.[17,18,22,23] Joint angles and linear and angular velocities and accelerations were calculated in the sagittal plane of the knee, as previously described.[17,18,22,23]

Raw EMG signals were full-wave rectified and smoothed using a 10 millisecond moving average window (linear enveloped) throughout the knee range of motion for each repetition.[17,18,22,23] EMG data were then normalized for each muscle as a percentage of each participant's MVIC, as previously described.[17,18,23,24] Normalized EMG data for the 3 repetitions (trials) were then averaged at corresponding knee angles between 0° and 100°.

A previously described[17,18,22,23,25] biomechanical model of the knee was used to continuously calculate patellofemoral joint forces throughout a 0° to 100° knee range of motion during the descent and ascent phases of the lunge. An overview of the biomechanical model is in the Appendix. After the joint forces were determined from the model, the maximum and minimum of the knee joint angle were then determined. Based on the maximum and minimum knee joint angle, the knee joint angles were chosen from odd number angles to get the averages of joint forces. We grouped all data points around knee angle α ($\alpha - 1$: $\alpha + 1$) during the same phase (ascending or descending) and used the averages of the group for this knee angle α.

A repeated-measures 2-way analysis of variance (ANOVA) was employed ($P < .05$) for each 10° knee angle (from 10° to 100°) during the lunge descent and ascent to assess the effects of step length (long step vs short step) and step height (landing at ground level vs landing up to a 10-cm height platform) on patellofemoral compressive force. Subsequently, the Holm–Bonferroni sequential correction was employed to adjust the significance level secondary to multiple ANOVAs being tested. Paired samples t tests with Bonferroni correction for multiple testing were used to assess pairwise comparisons among the different lunging conditions. Statistical analyses were not performed for patellofemoral joint stress values because stress values for each knee angle were derived from dividing force data by a constant. Therefore, any statistical difference between patellofemoral forces implies a difference between patellofemoral stresses.

Results

Tables 1 and 2 provide patellofemoral joint force (and stress) values, respectively. The P values shown in Tables 1 and 2 for the step length conditions represent the main effects of the ANOVA with the results of the Holm–Bonferroni sequential correction showing significant differences for P values less than or equal to .009. For 10° to 30° knee angles during the lunge descent and 10° to 40° knee angles during the lunge ascent, patellofemoral force (and stress) were significantly greater with a long step. In contrast, at 50° to 70° and 100° knee angles during the descent and 60° to 100° during the ascent, patellofemoral joint force (and stress) were significantly greater with a short step. For 50° to 100° knee angles during the lunge descent and 40° to 70° during the lunge ascent, patellofemoral joint force (and stress) were significantly greater at

Table 1 Mean (SD) Patellofemoral Joint Force (in Newton) Values While Performing the Forward Lunge With Step Length and Step Height Variations

	Step length variations				Step height variations			
Knee angle	Long-step lunge	Short-step lunge	*P*	*F*	Lunge at ground level	Lunge up to 10-cm platform	*P*	*F*
Lunge descent								
10°	64 (38)	41 (24)	.001*	15.4	55 (33)	49 (29)	.013	7.8
20°	113 (62)	79 (42)	.002*	13.1	96 (55)	95 (56)	.343	1.0
30°	148 (74)	105 (53)	.009*	9.8	125 (70)	128 (66)	.712	0.1
40°	167 (87)	135 (68)	.035	5.2	162 (89)	139 (65)	.063	3.9
50°	201 (115)	260 (152)	.009*	9.9	268 (145)	189 (109)	.005**	11.1
60°	297 (170)	470 (267)	.007*	10.8	421 (249)	340 (209)	.008**	10.3
70°	393 (230)	658 (394)	.002*	13.1	579 (348)	467 (286)	.007**	10.7
80°	521 (301)	773 (451)	.051	4.5	722 (419)	562 (329)	.006**	10.8
90°	619 (319)	822 (454)	.12	2.7	797 (406)	633 (385)	.007**	10.7
100°	615 (305)	839 (400)	.004*	11.8	815 (361)	646 (370)	.009**	10.0
Lunge ascent								
100°	762 (453)	1152 (398)	.002*	12.7	921 (424)	950 (483)	.669	0.2
90°	763 (448)	1300 (522)	<.001*	21.5	1021 (534)	1043 (605)	.673	0.2
80°	766 (447)	1278 (454)	.001*	16.6	1047 (526)	985 (530)	.381	0.8
70°	624 (354)	1069 (414)	<.001*	23.5	927 (468)	751 (400)	.008**	10.4
60°	491 (288)	822 (397)	<.001*	19.3	765 (406)	531 (315)	.002**	12.7
50°	367 (178)	432 (244)	.062	4.0	488 (237)	300 (129)	<.001**	20.2
40°	274 (142)	178 (105)	.002*	13.8	262 (152)	190 (102)	.001**	14.9
30°	204 (109)	107 (58)	<.001*	49.2	158 (91)	156 (91)	.647	0.2
20°	108 (59)	81 (49)	<.001*	20.6	88 (51)	102 (60)	.786	0.1
10°	65 (38)	39 (21)	<.001*	20.1	56 (35)	47 (28)	.092	3.2

Note: The mean values given for the 2 step length conditions (lunging with long step and lunging with short step) were collapsed across the 2 step height conditions (lunging at ground level and lunging up to 10-cm platform), and the mean values given for the 2 step height conditions were collapsed across the 2 step length conditions. The *P* values shown for step length conditions and step height conditions represent the main effects of the analysis of variance.

*Significant difference ($P < .009$) between step length conditions. **Significant difference ($P < .009$) between step height conditions.

Table 2 Mean (SD) Patellofemoral Joint Stress (MPa) Values While Performing the Forward Lunge With Step Length and Step Height Variations

	Step length variations				Step height variations			
Knee angle	Long-step lunge	Short-step lunge	*P*	*F*	Lunge at ground level	Lunge up to 10-cm platform	*P*	*F*
Lunge descent								
10°	0.37 (0.22)	0.24 (0.14)	.001*	15.4	0.32 (0.19)	0.28 (0.17)	.013	7.8
20°	0.61 (0.34)	0.43 (0.23)	.002*	13.1	0.52 (0.30)	0.52 (0.30)	.343	1.0
30°	0.61 (0.31)	0.43 (0.22)	.009*	9.8	0.52 (0.29)	0.53 (0.27)	.712	0.1
40°	0.58 (0.30)	0.46 (0.23)	.035	5.2	0.56 (0.31)	0.48 (0.23)	.063	3.9
50°	0.64 (0.37)	0.83 (0.49)	.009*	9.9	0.86 (0.46)	0.61 (0.35)	.005**	11.1
60°	0.86 (0.49)	1.35 (0.87)	.007*	10.8	1.21 (0.72)	0.98 (0.60)	.008**	10.3
70°	1.05 (0.62)	1.76 (1.06)	.002*	13.1	1.55 (0.93)	1.25 (0.76)	.007**	10.7
80°	1.30 (0.75)	1.93 (1.12)	.051	4.5	1.80 (1.04)	1.40 (0.83)	.006**	10.8
90°	1.44 (0.74)	1.92 (1.06)	.12	2.7	1.86 (0.95)	1.48 (0.90)	.007**	10.7
100°	1.35 (0.67)	1.84 (0.88)	.004*	11.8	1.78 (0.79)	1.41 (0.81)	.009**	10.0
Lunge ascent								
100°	1.67 (0.99)	2.52 (0.87)	.002*	12.7	2.02 (0.93)	2.08 (1.06)	.669	0.2
90°	1.78 (1.05)	3.03 (1.22)	<.001*	21.5	2.38 (1.25)	2.43 (1.41)	.673	0.2
80°	1.91 (1.11)	3.19 (1.13)	.001*	16.6	2.61 (1.31)	2.46 (1.32)	.381	0.8
70°	1.67 (0.95)	2.87 (1.11)	<.001*	23.5	2.49 (1.26)	2.01 (1.07)	.008**	10.4
60°	1.41 (0.83)	2.37 (1.14)	<.001*	19.3	2.20 (1.17)	1.53 (0.91)	.002**	12.7
50°	1.17 (0.57)	1.38 (0.78)	.062	4.0	1.56 (0.76)	0.96 (0.41)	<.001**	20.2
40°	0.94 (0.49)	0.61 (0.36)	.002*	13.8	0.90 (0.52)	0.66 (0.35)	.001**	14.9
30°	0.84 (0.45)	0.44 (0.24)	<.001*	49.2	0.66 (0.43)	0.64 (0.37)	.647	0.2
20°	0.59 (0.32)	0.44 (0.26)	<.001*	20.6	0.48 (0.27)	0.56 (0.33)	.786	0.1
10°	0.38 (0.22)	0.23 (0.12)	<.001*	20.1	0.33 (0.20)	0.27 (0.16)	.092	3.2

Abbreviations: ANOVA, analysis of variance; MPa, megapascal. Note: The mean values given for the 2 step length conditions (lunging with long step and lunging with short step) were collapsed across the 2 step height conditions (lunging at ground level and lunging up to 10-cm platform), and the mean values given for the 2 step height conditions were collapsed across the 2 step length conditions. The *P* values shown for step length conditions and step height conditions represent the main effects of the ANOVA.

*Significant difference ($P < .009$) between step length conditions. **Significant difference ($P < .009$) between step height conditions.

ground level compared with the forward lunge up to a 10-cm platform.

There were no significant interactions between lunge step length and step height conditions. During the lunge descent, *P*/*F* values for the interaction ranged between .102/3.091 and .814/0.058, and during the lunge ascent, *P*/*F* values for the interaction ranged between .107/2.996 and .706/0.149.

Descriptive data for patellofemoral joint force (and stress) are provided in Figures 2 and 3. Patellofemoral joint force (and stress) generally increased as knee flexion increased during the descent phase and decreased progressively as knee flexion decreased during the ascent phase. Moreover, for a given knee angle, patellofemoral joint force (and stress) were generally slightly greater during the ascent phases compared with the descent phases. Figure 2 visually depicts greater force (and stress) at lower knee angles for the long-step lunge and greater force (and stress) at higher knee angles for the short-step lunge. Figure 3 demonstrates greater patellofemoral compressive force (and stress) at higher knee angles lunging at ground level compared with lunging up a 10-cm platform.

Discussion

As hypothesized, patellofemoral compressive force (and stress) were greater while performing the forward lunge at ground level compared with up to a 10-cm platform; however, these differences were only significant at moderate to high knee angles. These data provide insights into how the patellofemoral joint can be loaded and progressed in PFPS rehabilitation as a function of knee angle and step height. There are no studies we are aware of in which patellofemoral joint loading during lunging with various step heights has been examined, in spite of these various lunging techniques commonly being employed in the clinic during PFPS rehabilitation. Very early in PFPS rehabilitation, performing "mini" lunges with lower knee angles (0°–40°) may be appropriate given that patellofemoral joint loading is relatively low; lunging at ground level or up to a 10-cm platform may both be appropriate given that there was no difference in patellofemoral joint loading between these 2 step height conditions at these lower knee angles. As lunging progression moves beyond 40° knee angle, patellofemoral joint force (and stress) progressively increase. Furthermore,

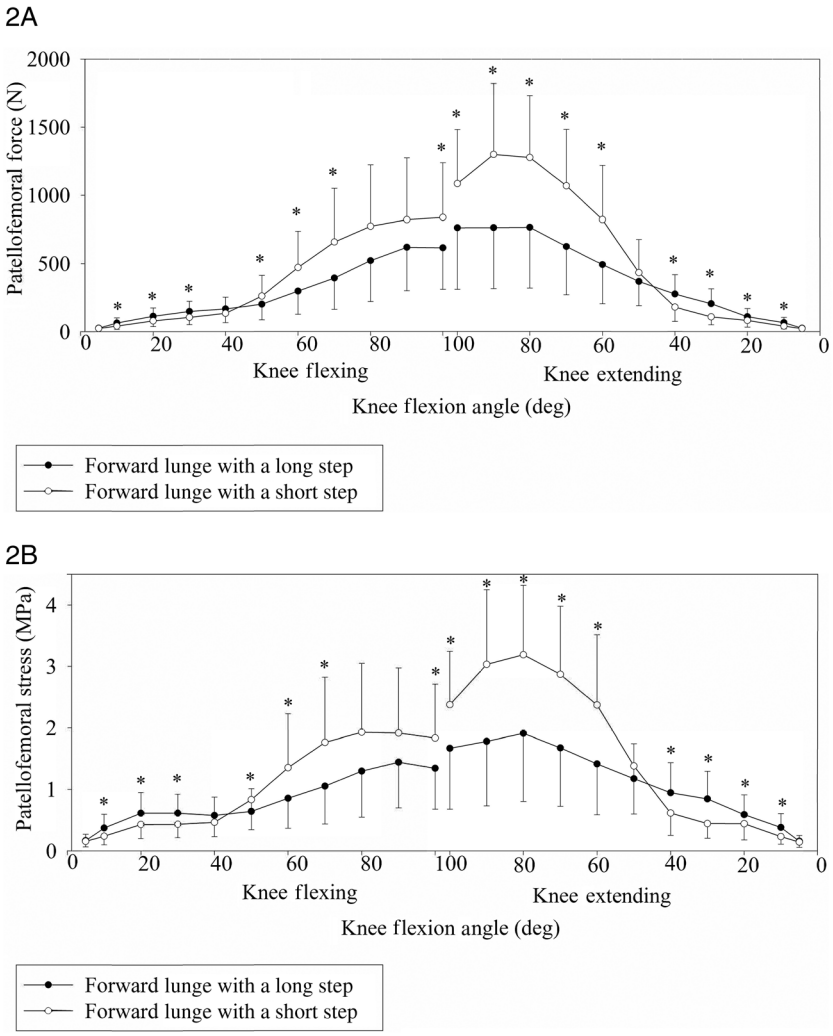


Figure 2 — (A) Mean (SD) patellofemoral compressive force in lead (right) knee for forward lunge with a long step and forward lunge with a short step. (B) Mean (SD) patellofemoral stress in lead (right) knee for forward lunge with a long step and forward lunge with a short step. *Significant difference ($P < .009$) between long and short step lengths.

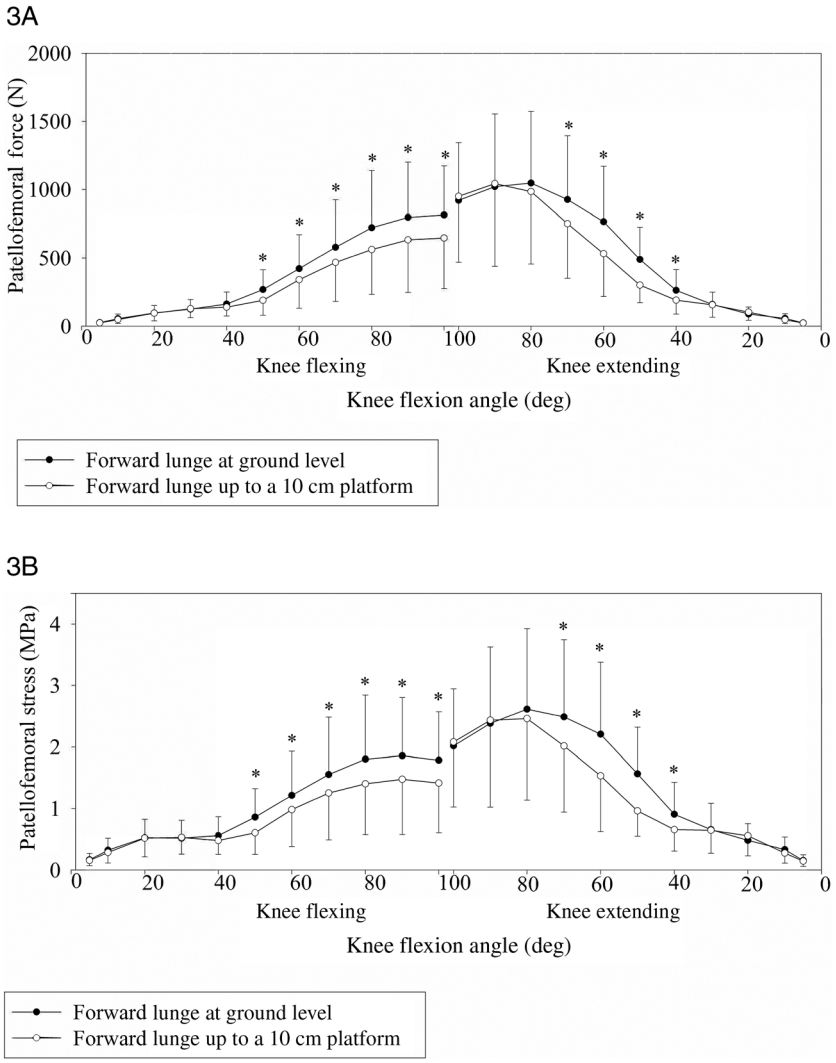


Figure 3 — (A) Mean (SD) patellofemoral compressive force in lead (right) knee for forward lunge at ground level and forward lunge up to a 10-cm platform. (B) Mean (SD) patellofemoral stress in lead (right) knee for forward lunge at ground level and forward lunge up to a 10-cm platform. *Significant difference ($P < .009$) between long and short step lengths.

joint force (and stress) were greater at ground level. Therefore, lunging progression as a function of knee angle and step height during PFPS rehabilitation can proceed from lunging at more functional lower knee angles (eg, 0°–50° knee angles) to lunging at higher knee angles (eg, 60°–100° knee angles) up to a 10-cm platform, followed by lunging at higher knee angles at ground level.

Patellofemoral joint loading (compressive force and stress) during bodyweight-only lunging has never been examined before between the short and long step forward lunge, in spite of it commonly occurring in the clinic during the early stages of PFPS rehabilitation. Escamilla and colleagues[18] did examine patellofemoral joint loading during the forward lunge with short step and long step, but these authors used a 12-repetition maximum
(12 RM) external load, which is more appropriate in the latter stages of PFPS rehabilitation and strength training. In contrast, the bodyweight lunge is more appropriate earlier in PFPS rehabilitation. As hypothesized, patellofemoral joint force (and stress) were both greater while performing the bodyweight forward lunge with a short step compared with a long step, particularly at higher knee angles. In contrast to our hypothesis, at lower knee angles between 0° and 40°, patellofemoral joint force (and stress) were greater while performing the forward lunge with a long step compared with short step. However, it should be noted that patellofemoral force (and stress) are relatively low at knee angles between 0° and 40°, and although statistically significant, they may not be clinically relevant. Therefore, patellofemoral joint loading during the forward lunge is greater with a long step at lower knee angles but greater during a short step at higher knee angles. Nevertheless, when the goal is to minimize patellofemoral joint force (and stress), the forward lunge with a long step may be preferred compared with the forward lunge with a short step, especially given that patellofemoral joint force (and stress) increase as knee angle increases up to approximately 80° to 90° knee angles.

Like the current study, Escamilla et al[18] also reported significantly greater patellofemoral joint force (and stress) at higher knee angles (70°–90°) for the forward lunge with a short step compared with a forward lunge with a long step. In contrast, they did not find

greater patellofemoral joint force (and stress) at lower knee angles for the forward lunge with a long step compared with a forward lunge with a short step. Given that the step length variations were similar between studies, this implies that using or not using external resistance during the forward lunge may result in some different findings.

Comparing results from Escamilla et al[18] to the current study, patellofemoral joint force (and stress) values were 2 to 3 times greater in forward lunges with a 12 RM external load compared with the values in the current study with no external loads added to bodyweight. Interestingly, the magnitudes of patellofemoral joint loading from Hofmann and colleagues,[19] who also examined the bodyweight lunge, were more similar to patellofemoral joint loading of the 12 RM lunging intensity employed by Escamilla et al[18] than the bodyweight lunge in the current study. It is likely that these differences in patellofemoral joint loading between these studies are due to methodological differences.

Hofmann et al[19] was the first study to examine patellofemoral joint loading for both the lead and trail limbs during the forward lunge and reported similar loading between the 2 limbs. Like the current study, Hofmann et al[19] reported greater patellofemoral joint compressive force (and stress) during the lunge when the shank was inclined forward (ie, knee forward beyond the toes) compared with when the shank was vertical. During the forward lunge with a short step in the current study, it was visually observed that the lead knee first began translating beyond the distal toes at approximately 60° knee angle, which corresponds to the knee angle where patellofemoral joint force (and stress) in the lunge with a short step surpass the loads with a long step. It has been stated for some time in rehabilitation, do not let your toes go over your knee. However, during the forward lunge with a short step, the lead knee continued translating beyond the toes as the lead knee continued flexing (Figure 1A and 1C), translating 7.1 (3.5) cm beyond the distal toes at maximum lead knee flexion. In contrast, the lead knee did not translate beyond the toes throughout the knee range of motion during the forward lunge with a long step because the shank was relatively vertical (Figure 1B and 1D). Because significantly greater patellofemoral joint force (and stress) at higher knee angles occurred with a short step, there appears to be a relationship between anterior knee translation and increased patellofemoral joint force (and stress). This supports previous findings from Escamilla et al.[18]

The primary cause of the greater patellofemoral joint force (and stress) at higher knee angles is greater quadriceps force at higher knee angles when lunging with both a short step and a long step.[18] At higher knee angles, the estimated quadriceps forces (calculated from the model) were approximately 20% to 30% greater for the forward lunge with a short step compared with a long step, whereas the estimated hamstring forces (calculated from the model) were approximately 25% to 35% greater with a long step compared with a short step.[18] Because patellofemoral force increases as quadriceps force increases, this helps explain why at higher knee angles patellofemoral force was greater lunging with a short step than lunging with a long step.

Using the forward lunge with a short step and high knee flexion may be problematic for patients with PFPS due to increased patellofemoral joint force (and stress). Our results support the belief of many clinicians and trainers that anterior lead knee translation beyond the toes during the forward lunge may be harmful to the patellofemoral joint and should be avoided. The relationship between increased anterior knee translation and increased patellofemoral joint force (and stress) should be examined during other weight-bearing exercises, such as the squat, because different technique variations can result in lesser or greater amounts of anterior knee translation.

Patellofemoral joint force (and stress) curves were similar in shape to each other. One exception was at higher knee angles between 80° and 100° in which patellofemoral joint stress began to plateau or decrease. This occurred because although patellar contact area increased nearly linearly between 80° and 100°, patellofemoral joint force began to plateau around 80° in the current study (then decrease from 90° to 100°) and around 70° to 80° in numerous squat and lunge studies in the literature that quantified patellofemoral compressive force (and stress).[18,19,22–24,26] Therefore, injury risk to the patellofemoral joint may not increase when training at higher knee angles (100°) compared with slightly lower knee angles (80°) because patellofemoral joint stress generally decreased between 80° and 100° knee angles (Table 2). One may also derive additional benefit from training at a higher knee angle (100°) compared with slightly lower knee angles (80°) because quadriceps, hamstrings, and gastrocnemius activity are generally greater at 100° knee angle compared with 80° knee angle.[18,19,22–24,26]

Mean peak patellofemoral joint force (and stress) magnitudes from the current study (1300 N and 3.19 MPa, respectively) for the bodyweight lunge are less than some weight-bearing exercises that use external weight, such as the 12 RM weighted barbell squat and leg press (4500–4700 N and 11–12 MPa),[23,27] 12 RM wall squat and one leg squat (approximately 3500 N and 9 MPa),[22] and barbell squat using a 35% bodyweight external load (approximately 2400 N and 13 MPa),[28] less than some weight-bearing exercises that use body weight only, such as the bodyweight squat (approximately 1700 N and 9.3 MPa)[28] and going up and down stairs (approximately 2500 N and 7 MPa),[29] but more than other weight-bearing functional activities, such as fast walking (approximately 900 N and 3.13 MPa).[30]

Unfortunately, it is currently unknown what patellofemoral joint force or stress magnitudes, and over what time duration, can ultimately lead to patellofemoral pathology. There are many factors that may contribute to patellofemoral pathology, such as overuse or trauma, dysfunctional extensor mechanism, weakness in the quadriceps or hip external rotators, tight quadriceps, hamstrings, or iliotibial band, lower-extremity malalignment, and excessive rear foot pronation. Nevertheless, clinicians can use information regarding patellofemoral joint force (and stress) magnitudes among different weight-bearing exercises, technique variations, and functional activities to be able to make informed decisions regarding which exercise they choose to employ during patellofemoral rehabilitation.

There are limitations in the current study. First, magnetic resonance imaging (MRI) knee kinematic data have shown during the weight-bearing squat that the femur moves and rotates underneath a relatively stationary patella, and excessive femoral rotation may increase patellofemoral joint stress on the contralateral patellar facets.[15,16,31] Unfortunately, MRI knee kinematic data do not currently exist while performing the forward lunge exercise. Therefore, it is unknown how much femoral rotation occurs during the forward lunge and how this rotation varies among healthy individuals and those with pathologies.

Second, muscle and patellofemoral joint forces were estimated from modeling techniques and not measured directly, which is currently not possible in vivo in healthy subjects. However, the EMG-driven model matched well (less than 2% differences) with the torques generated from the inverse dynamics. Another

limitation is that patellofemoral joint stress magnitudes were measured using patellar contact area values from MRI data from the literature and were not measured directly. However, the contact areas used from the literature were determined during loaded weight-bearing exercise in healthy male and female subjects, similar to the current study. Moreover, the near linear and direct relationship between contact area and knee angle has been shown to be similar among studies.[13,32–35] This implies that the patellofemoral joint stress curve patterns shown in Figures 2B and 3B using contact areas from the literature will be similar to patellofemoral joint stress curve patterns if contact areas were measured directly using MRI. It is important for clinicians to understand patellofemoral joint stress patterns during lunging in order to properly prescribe different types of lunging exercises during knee rehabilitation.

Patellofemoral joint loading during lunging changes according to step length, step height, and knee angle. When the goal is to initially minimize patellofemoral joint force (and stress) and then gradually progress loading, it may be prudent to perform forward lunge: (1) initially, with minimal knee flexion (0°–30°); (2) then, with moderate knee flexion (0°–60°); (3) with long step and deep knee flexion (0°–100°) up to a 10-cm platform; and (4) finally, with long step and deep knee flexion (0°–100°) at ground level.

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Appendix: Biomechanical Model

A previously described[17,18,25] biomechanical model of the knee was used to continuously calculate patellofemoral joint forces throughout a 0° to 100° knee range of motion during the lunge descent and lunge ascent. Resultant force and torque equilibrium equations were calculated using inverse dynamics and the biomechanical knee model.[17,18,25] Moment arms of muscle forces and angles of the line of action for muscles were represented as polynomial functions of the knee flexion angle.[17,18,25]

Quadriceps, hamstrings, and gastrocnemius muscle forces were calculated as previously described.[17,18,25] Because the accuracy of calculating muscle forces depends on accurate calculations of a muscle's physiological cross-sectional area (PCSA), maximum voluntary contraction force per unit PCSA, and the EMG–force relationship, resultant force and torque equilibrium equations may not be satisfied. Therefore, each muscle force $F_{m(i)}$ was modified by the following equation at each knee angle[17,18,25]:

$$F_{\mathrm{m}(i)} = \mathrm{c}_i \mathrm{k}_{li} \mathrm{k}_{\mathrm{v}i} \mathrm{A}_i \sigma_{\mathrm{m}(i)} [\mathrm{EMG}_i / \mathrm{MVIC}_i],$$

where A_i was PCSA of the ith muscle examined, $\sigma_{\mathrm{m}(i)}$ was MVIC force per unit PCSA of the ith muscle, EMG_i and MVIC_i were EMG window averages of the ith muscle EMG during exercise and MVIC trials, c_i was a weight factor adjusted in a computer optimization program to minimize the difference between the resultant torque from the inverse dynamics (T_{res}) and the resultant torque calculation from the biomechanical model ($T_{\mathrm{m}i}$), k_{li} represented each muscle's force–length relationship as a function of hip and knee angles (based on muscle length, fiber length, sarcomere length, pennation angle, and cross-sectional area),[17,18,25] and k_{vi} represented each muscle's force–velocity relationship based on a Hill-type model for eccentric and concentric muscle actions using 2 equations from Zajac[36] and Epstein and Herzog[37]:

$$(1)\ \text{Concentric}: k_v = (b - (a/F_{\mathrm{o}})v)/(b + v) \text{ and}$$

$$(2)\ \text{Eccentric}: k_v = C - (C-1)(b + (a/F_{\mathrm{o}})v/(b - v),$$

with F_0 representing isometric muscle force; l_0 = muscle fiber length at rest; v = velocity; $a = 0.32\ F_0$; $b = 3.2\ l_0$/s; and $C = 1.8$. Forces generated by the knee flexors and extensors at MVIC were assumed to be linearly proportional to their PCSA. Muscle force per unit physiological cross-sectional area at MVIC was 35 N/cm^2 for the knee flexors and 40 N/cm^2 for the quadriceps.[38–40]

The objective function used to determine each ith muscle's coefficient c_i was as follows:

$$\min f(c_i) = \sum_{i=1}^{n_m} (1 - c_i)^2 + \lambda \left(T_{\mathrm{res}} - \sum_{i=1}^{n_m} T_{\mathrm{mi}} \right)^2,$$

subject to $c_{\mathrm{low}} \le c_i \le c_{\mathrm{high}}$, where c_{low} and c_{high} were lower and upper limits for c_i, and λ was a constant. The weight factor "c" was to adjust the final muscle force calculation. The bounds on "c" were set between 0.5 and 1.5. The assumptions associated with this model were that knee torques from cruciate and collateral ligament forces and bony contact were assumed to be negligible, as were forces and torques out of the sagittal plane. Patellofemoral joint force was a function of patellar tendon force and quadriceps tendon force. Patellar tendon force was calculated by the quadriceps tendon force and the ratio of the patellar tendon force and the quadriceps tendon force, as previously described.[17,25,41] The angles between the patellar tendon, quadriceps tendon, and patellofemoral joint were expressed as functions of knee angle.[17,25,41]

Patellofemoral joint stress, which was calculated every 10° between 0° and 100° knee angles, was expressed as the ratio of patellofemoral joint force (calculated from the biomechanical model described earlier)[17,18,25] and patellar contact area. Patellar contact areas were determined for 10° intervals between 0° and 100° knee angles. Contact areas from in vivo MRI data from Salsich et al,[42] who also used both male and female participants with healthy knees and had them perform weight-bearing exercise, were employed for 0° (146 mm^2), 20° (184 mm^2), 40° (290 mm^2), and 60° (347 mm^2) knee angles. These 4 contact area values formed a near linear relationship as a function of knee angle, resulting in a line of best fit equation of $y = 3.55x + 135$ ($r = .98$) with y = contact area and x = knee angle. This line of best fit equation was used to determine contact areas for 10° knee angle (171 mm^2), 30° knee angle (24 mm^2), and 50° knee angle (313 mm^2). The contact areas for 40°, 50°, and 60° knee angles were used to develop the line of best fit equation $y = 2.81x + 176$ ($r = .99$), which was used to determine contact areas for 70° knee angle (373 mm^2), 80° knee angle (401 mm^2), 90° knee angle (429 mm^2), and 100° knee angle (457 mm^2). Like the current study, a near linear relationship between patellar contact area and knee angles has been reported between approximately 0° and 100° knee angles in studies involving weight-bearing exercises.[13,34,42]

All biomechanical models have limitations. Muscle and patellofemoral joint forces were estimated from modeling techniques and not measured directly, which is currently not possible in vivo in healthy participants. However, using numerical boundaries between 0.5 and 1.5 for the term "c" in the muscle force equation $F_{\mathrm{m}(i)} = c_i k_{li} k_{vi} A_i \sigma_{m(i)} [\mathrm{EMG}_i / \mathrm{MVIC}_i]$ resulted in the torques predicted by the EMG-driven model matching well (less than 2% differences) with the torques generated from the inverse dynamics. The "c" term was used as a boundary for adjusting each muscle force and was determined by the optimization program for each knee angle. The cost function had a component to force c to be as close to 1.0 (optimal) as possible, making the model more sensitive to the model parameters and minimizing the amount of error.

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